

PROJECT DESIGN PHASE

PROPOSED SOLUTION

Date	30 September 2026
Team ID	SWTID-2026-3024
Project Name	AI-Augmented Backend Development AI BlogNest API
Maximum Marks	2 Marks

Proposed Solution :

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional blog content management involves considerable manual effort for creating, organizing, updating, and summarizing content. Managing multiple blog posts, users, comments, and access permissions can also become difficult. There is a need for a centralized system that can simplify blog management while providing AI assistance for content creation.
2.	Idea / Solution description	AI BlogNest is an AI-powered blog content management system developed as a secure RESTful backend. It provides APIs for user authentication, role-based access, blog management, comment management, database operations, and AI-assisted content generation and summarization. The system uses Node.js and Express.js for backend development, MongoDB with Mongoose for data management, JWT for authentication, and bcrypt.js for password protection.
3.	Novelty / Uniqueness	The uniqueness of AI BlogNest is the integration of Artificial Intelligence with a centralized blog management backend. Instead of depending completely on manual content creation, users can use AI assistance to generate blog content and summarize existing content.

		The system also combines AI functionality with secure authentication, role-based access, database management, and RESTful APIs in a single platform.
4.	Social Impact / Customer Satisfaction	AI BlogNest can reduce the time and effort required for digital content creation and management. AI-assisted generation can help content creators prepare initial content more efficiently, while summarization can help users understand lengthy information quickly. Centralized management and secure role-based access also provide users with a more organized and controlled content management experience.
5.	Business Model (Revenue Model)	The system can follow a subscription-based model where users can access different levels of features based on their selected plan. Possible revenue sources include: • Free basic plan with limited AI usage. • Premium subscription for increased AI content generation. • Professional plans for advanced content management features. • Business or organizational plans for multiple users. • Additional paid AI usage beyond the included limits.
6.	Scalability of the Solution	AI BlogNest is designed using a modular backend architecture, which allows additional features and services to be added as the system grows. The solution can be scaled by: • Supporting a larger number of users and blog posts. • Adding advanced search and filtering.

		• Introducing multilingual content generation. • Adding content recommendation features. • Supporting media and file management. • Adding analytics and dashboards. • Deploying the application on cloud infrastructure. • Integrating additional AI services and models.
--	--	---